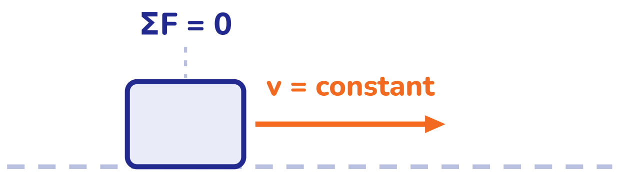
Newton's Laws of Motion

Quick Revision Notes · Theory & Standard Results · JEE Main / Advanced · NEET

The Three Laws

First Law — Inertia

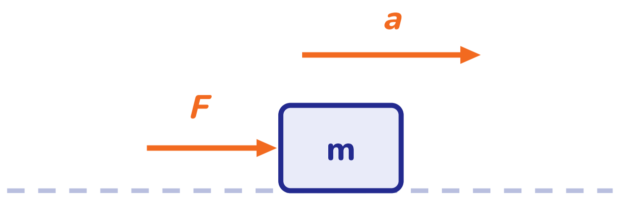


A body stays at rest or in uniform straight-line motion unless a **net external force** acts on it.

Defines **inertial frames**. Mass is the measure of inertia.

Force changes motion — it is not needed to *maintain* it.

Second Law



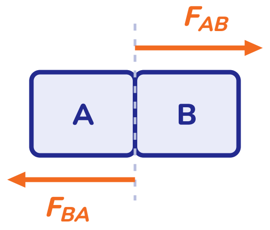
$$\vec{F}_{net} = \frac{d\vec{p}}{dt}$$

KEY

$$\vec{F}_{net} = m\vec{a} \text{ (constant mass)}$$

Valid in inertial frames; $\vec{a}$ is instantaneous and along $\vec{F}_{net}$; $\vec{p} = m\vec{v}$.

Third Law



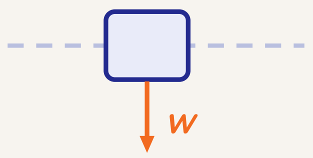
$$\vec{F}_{AB} = -\vec{F}_{BA}$$

Equal, opposite, **simultaneous** — and acting on **different bodies**, so they never cancel each other.

Action and reaction never appear in the same free body diagram.

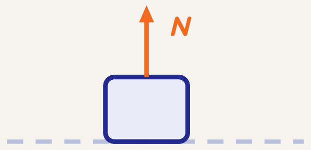
Common Forces & Free Body Diagrams

Common Forces



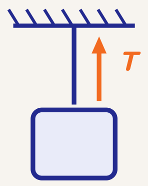
Weight

$W = mg$, vertically down



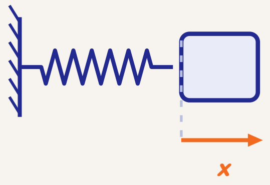
Normal

$\perp$ to surface, push only



Tension

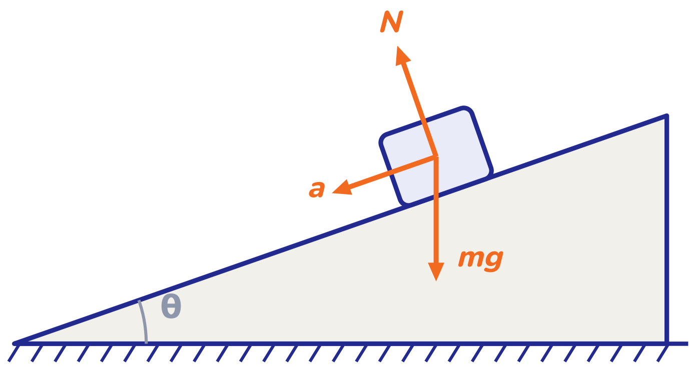
along string, pull only; same throughout an ideal string



Spring

$F = -kx$ (restoring)

How to Draw an FBD



1. Isolate the body
2. Mark every **external** force at its point of application
3. Choose axes along & perpendicular to $\vec{a}$
4. Write $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$

One diagram per body — never mix two bodies in the same FBD.

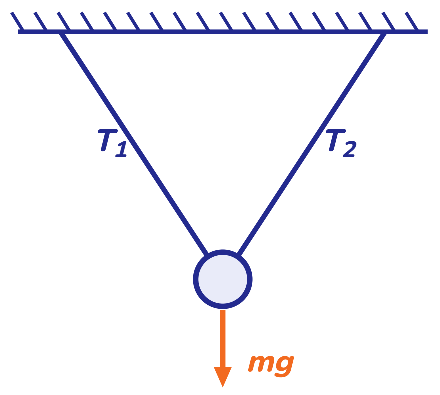
Trap: $N = mg$ is **not a rule** — always solve for N; it changes with acceleration and applied forces.

Equilibrium

Condition

$$\Sigma\vec{F} = 0$$

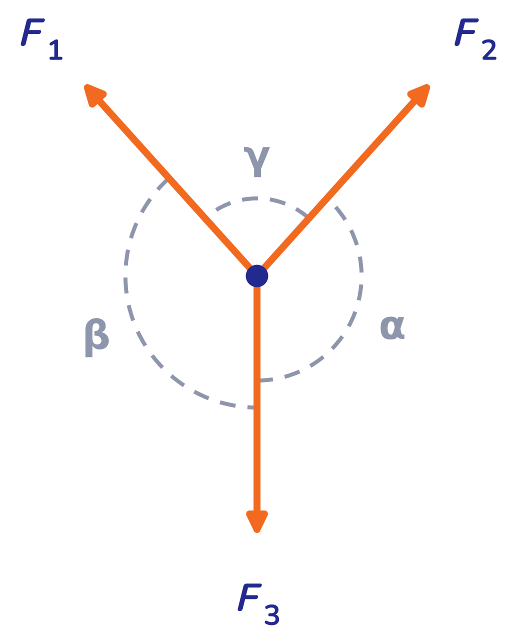
Body remains at rest or moves with **uniform velocity**. Holds componentwise: $\Sigma F_x = 0, \Sigma F_y = 0$.



A bob held by two inclined strings — three concurrent forces T_1, T_2, mg in equilibrium. Solve with Lami's theorem.

Resolve along x and y, or use the sine rule below — same answer.

Lami's Theorem



Each angle sits opposite its force: α opposite F_1, β opposite F_2, γ opposite F_3.

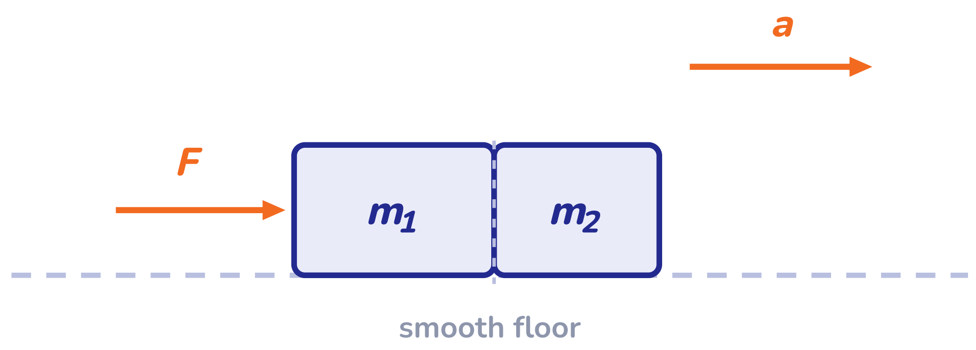
KEY

$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

Only for **three concurrent** coplanar forces in equilibrium.

Blocks in Contact & Connected by Strings

Blocks in Contact



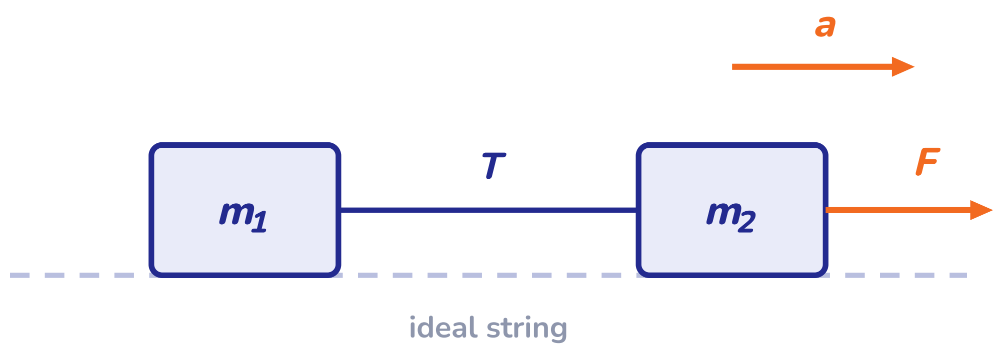
$$a = \frac{F}{m_1 + m_2}$$

KEY

$$N = \frac{m_2 F}{m_1 + m_2}$$

Contact force on m_2 = force needed to accelerate m_2 alone.

Blocks Joined by a String



$$a = \frac{F}{m_1 + m_2}$$

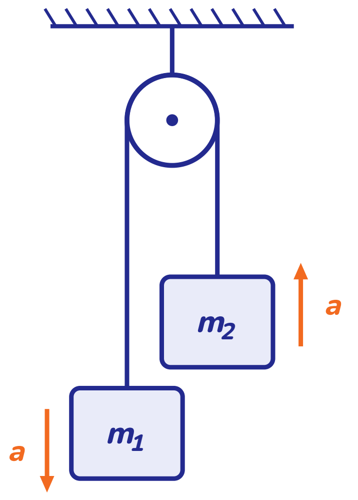
KEY

$$T = \frac{m_1 F}{m_1 + m_2}$$

Rule: tension at any section = (total mass dragged **behind** it) × a.

Pulley Systems — Standard Results

Atwood Machine $(m_1 > m_2)$

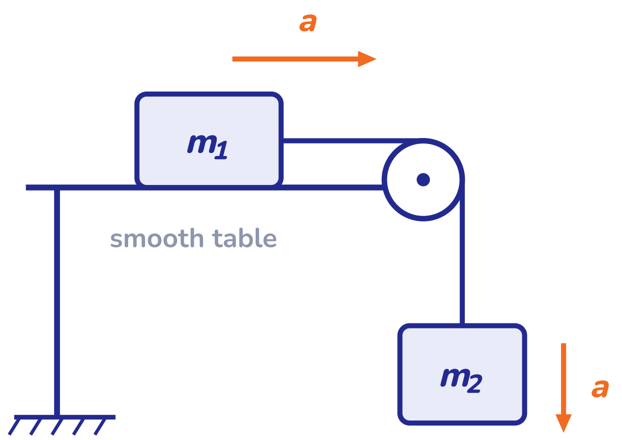


KEY

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} \qquad T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

Force on the pulley clamp $= 2T$

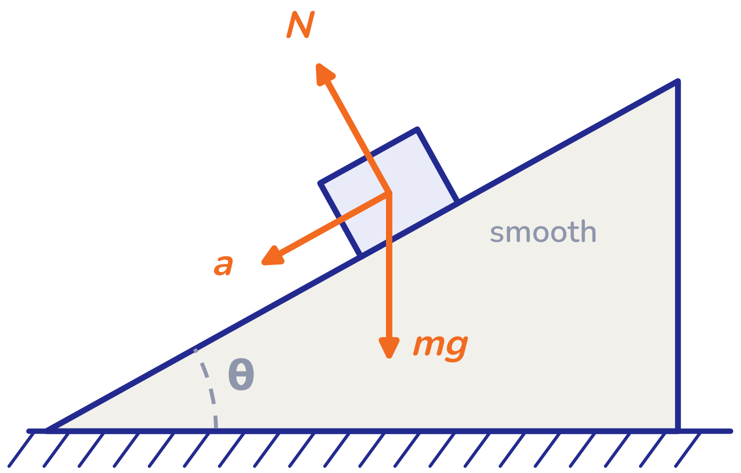
Table + Hanging Mass



$$a = \frac{m_2 g}{m_1 + m_2} \qquad T = \frac{m_1 m_2 g}{m_1 + m_2}$$

Smooth table, ideal pulley & string

Smooth Incline (θ)



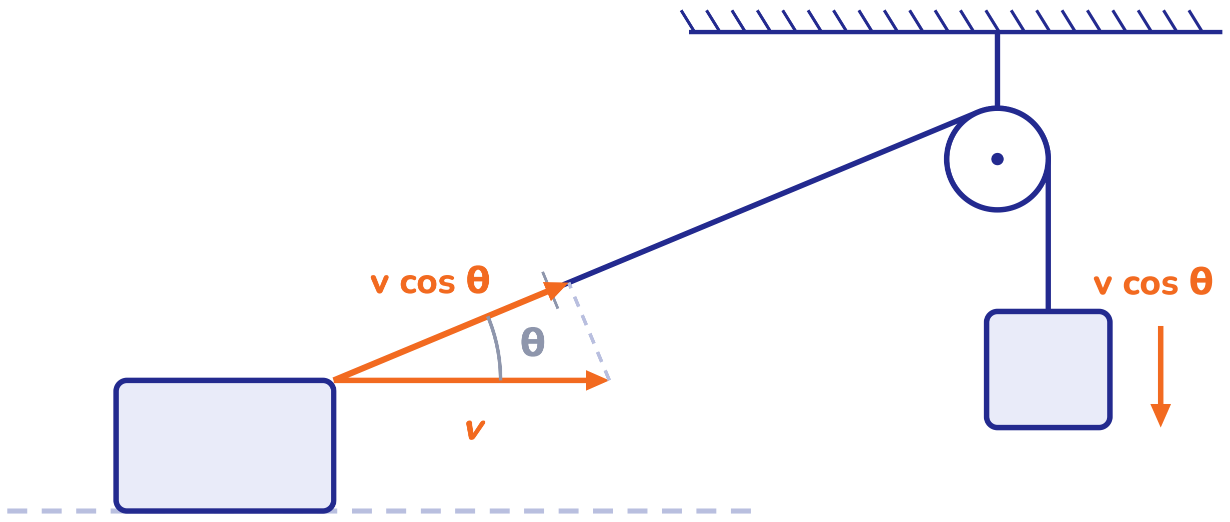
$$a = g\sin\theta \text{ (down the incline)}$$

$$N = mg\cos\theta$$

Independent of mass — every block slides the same way.

Constraint Relations

String Constraint



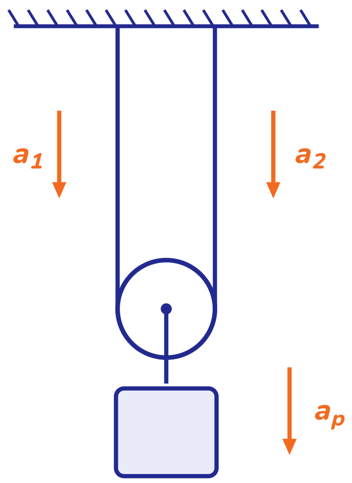
Total string length is **constant** ⇒ sum of velocity components **along the string** is zero.

Or: write the length equation and **differentiate twice**.

SHORTCUT

$\Sigma \vec{T} \cdot \vec{a} = 0$ over all string segments

Standard Results



$$a_{pulley} = \frac{a_1 + a_2}{2}$$

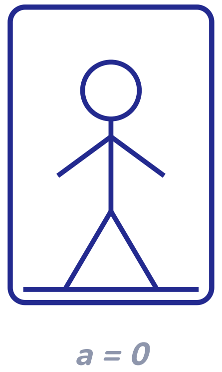
String over a **movable pulley** — the pulley moves with the average of the two end accelerations.

Fixed pulley: both ends move with equal speeds. Movable pulley: they need not.

Wedge constraint: two bodies in contact have equal velocity components along the **common normal**.

Apparent Weight in a Lift

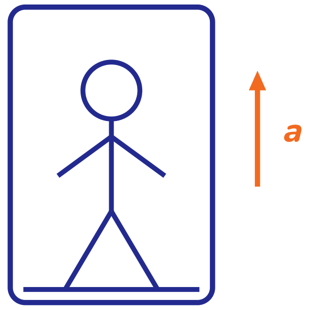
APPLICATIONS



At rest / uniform velocity

$$N = mg$$

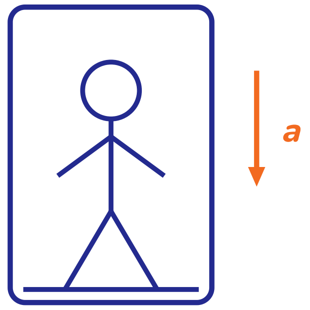
true weight



Accelerating **up**

$$N = m(g + a)$$

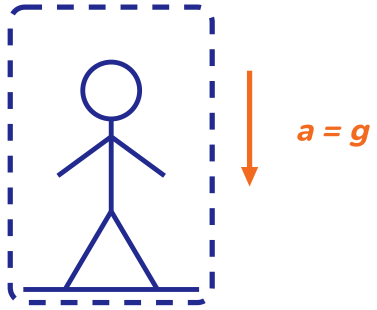
feels heavier



Accelerating **down**

$$N = m(g - a)$$

feels lighter



Free fall

$$N = 0$$

weightlessness

KEY

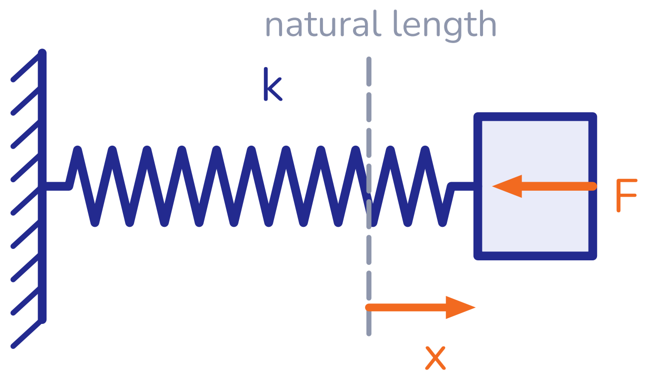
$$N = m(g \pm a)$$

Note: if downward $a > g$, the person loses contact with the floor (N cannot be negative).

Spring Force

Hooke's Law

$$F = -kx$$



A spring exerts force only through its **extension or compression** x; the force always opposes the deformation.

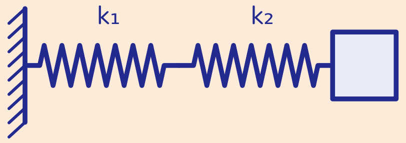
Trap: just after a cut/release, **spring force cannot change instantly** (length can't jump) — but **string tension can**.

Combinations & Cutting

KEY

Series

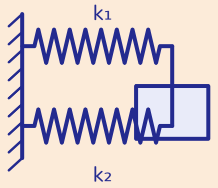
$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$



KEY

Parallel

$$k_{eq} = k_1 + k_2$$



Cutting $k \propto \dfrac{1}{\ell} \Rightarrow n$ equal parts: each $k\backslash' = nk$

Shorter spring = stiffer spring — half the length, twice the stiffness.

Pseudo Force — Non-Inertial Frames

The Rule

In a frame accelerating with $\vec{a}_0$, add on **every mass**:

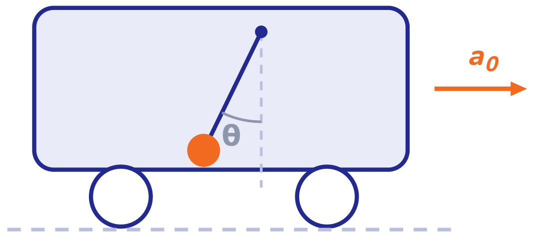
KEY

$$\vec{F}_{pseudo} = -m\vec{a}_0$$

...then apply Newton's laws as usual.

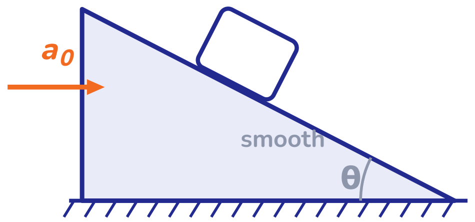
Note: a pseudo force is not a real interaction — it has **no reaction pair**.

Classic Results



$$\tan\theta = \frac{a_0}{g}$$

Pendulum tilts backwards in an accelerating car



$$a_0 = g\tan\theta$$

Block stays put on the wedge when pushed with this a_0

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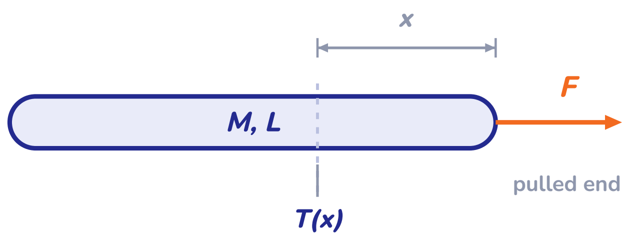
The Power of Real Gurus

Massive Ropes & Common Traps

MISC

Rope with Mass

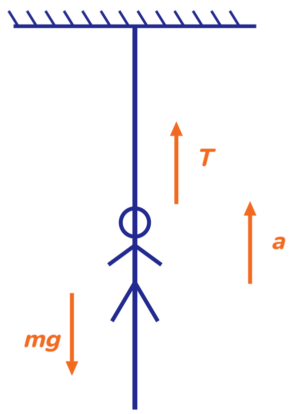


KEY

$$T(x) = F\left(1 - \frac{x}{L}\right)$$

x measured from the **pulled end**; tension falls linearly to zero at the free end.

Climbing a Rope



Up with a: $T = m(g+a)$

Uniform speed: $T = mg$

Rope breaks if $T > T_{max}$

Same result for a monkey or a man — only $\vec{a}$ matters.

Traps Checklist

1. Never assume $N = mg$
2. Tension differs across a **massive** string or pulley
3. Action–reaction forces act on **different bodies**
4. Write **constraint relations** before $F = ma$

Check the frame first — inertial or not?